SlotSync

Complete Product Blueprint

Updated edition: Customer History + Appointment Reminder Notifications

Product promise

Wherever a customer discovers a business, they can book directly—and the business manages every appointment, customer relationship, and reminder in one place.

Build principle

Deliver a polished, dependable daily-use booking tool for local service businesses. Prioritize the complete booking journey, customer history, reliable reminders, and a clean owner dashboard over a broad but unfinished feature set.

Hackathon build target • Cloudflare-first architecture • Mobile-first responsive UI

1. Executive summary

SlotSync solves a common operational problem: local service providers receive appointments through WhatsApp, phone calls, social media, and walk-ins, but those bookings often end up scattered across chats, notebooks, and calendars. SlotSync gives each business a direct booking page and one central schedule.

The product is designed for salons, clinics, tutors, mechanics, photographers, home-service providers, fitness professionals, consultants, and other appointment-based businesses. It is one configurable platform with industry presets—not a collection of unrelated apps.

The updated scope makes two capabilities first-class: (1) a customer profile with a complete appointment timeline, and (2) configurable appointment reminder notifications to reduce forgotten bookings and make changes easier to communicate.

2. Users, problems, and core journeys

Business owner/staff: publishes services and availability, shares a booking link, adds appointments received by phone or walk-in, reviews customer history, and monitors reminders.

Customer: opens a link without installing an app, selects a service and available slot, receives a confirmation and reminders through supported channels, and can view rescheduling/cancellation instructions.

Core journey: discover business → select service → choose slot → submit contact details → server confirms slot → confirmation message → reminder(s) → appointment completion → customer history updated → optional review/rebooking.

3. Complete feature catalogue

A. Business profile and setup

• Business registration, sign-in, and secure session management.

• Industry presets with configurable services, durations, buffers, prices, staff/resources, working hours, breaks, holidays, and service areas.

• Business name, logo, cover image, description, contact details, location, accessibility information, and public booking URL.

• Business-specific timezone and clear booking/cancellation policies.

B. Customer booking

• Public, mobile-friendly booking page; customers do not need to install SlotSync.

• Service and optional staff/resource selection, live availability, contact form, optional booking note, and clear price/duration.

• Instant confirmation or owner approval, depending on business settings.

• Reschedule/cancel flow, booking reference, waitlist, alternative slots, and safe handling of simultaneous booking attempts.

C. Owner command center

• Day/week/agenda views; filters by staff, service, and status.

• Quick Add for calls, WhatsApp requests, and walk-ins; label booking source.

• Prevent overlapping bookings and show alternatives when a slot is unavailable.

• Needs Attention panel for pending approvals, cancellations, reminder failures, and waitlist opportunities.

• Customer search, history access from bookings, CSV export, and audit trail.

D. Customer history and relationship management

• A business-scoped customer directory searchable by name or normalized phone number.

• Customer profile: name, phone, optional email, customer-since date, upcoming booking, last visit, total bookings, completed visits, cancellations/no-shows, and preferred services/staff where appropriate.

• Chronological appointment timeline containing date/time, service, assigned staff, booking source, status, amount if relevant, and permitted staff notes.

• Returning-customer recognition: match an existing customer within the same business when the customer rebooks with the same phone number; avoid silent merging where identity is uncertain.

• Quick rebooking from a past appointment, with current availability and price revalidated.

• Duplicate detection and a controlled merge workflow; preserve appointment links and audit changes.

• Role-based access to notes and customer details; export and deletion-request workflow.

• Industry-specific data minimization. For example, do not put sensitive clinical information in generic notes.

E. Reminder messages and customer notifications

• Send a booking confirmation after the server has successfully saved the booking.

• Configurable reminder schedule, for example 24 hours and 2 hours before the appointment; business owners can enable, disable, or adjust supported reminder timings.

• Include business name, service, local date/time and timezone, location or visit instructions, booking reference, and a clear reschedule/cancel path.

• Notify customers when the business approves, reschedules, or cancels a booking; notify about waitlist slot offers where enabled.

• Support in-app/email/SMS/WhatsApp only when the relevant channel and provider are actually configured. Never claim a message was sent merely because a reminder job was queued.

• WhatsApp share links may prefill a message for a person to send; automated WhatsApp notifications require an approved integration and applicable consent/template rules.

• Track notification state: queued, sent/accepted by provider, delivered/read when the provider exposes it, failed, retry scheduled, and permanently failed.

• Prevent duplicate sends with idempotency keys; retry temporary failures with backoff; stop or update stale reminders after a reschedule or cancellation.

• Customer consent and contact preferences, opt-out handling where applicable, quiet-hour rules, rate limits, and a manual resend action.

• Owner-facing reminder health panel: upcoming reminders, failures, retries, and a safe fallback action such as copy message or open a prefilled WhatsApp link.

F. Ratings and reputation

• 1–5 star rating and optional written review.

• Verified-booking marker only for eligible completed appointments; one review per booking.

• Owner replies, report/moderation flow, and transparent empty-rating state.

• Industry-specific rating categories only where useful; do not fabricate reviews or ratings.

G. Distribution and discovery

• Permanent booking URL and downloadable QR code.

• Share templates for WhatsApp, Instagram bio, website Book Now button/embed, and supported Google Business Profile appointment/website fields.

• Track source tags such as website, Instagram, WhatsApp, call, and walk-in where captured.

H. Analytics and daily operations

• Today/upcoming bookings, available slots, utilization, completions, cancellations, and no-shows.

• Expected versus collected revenue only when relevant amounts are recorded; label estimates clearly.

• Repeat-customer rate, popular services, busy periods, booking source performance, and end-of-day summary.

• Avoid presenting sample or invented numbers as live business data.

4. Industry presets

Industry	Useful configurable details
Salon/barber	Staff preference, service combinations, buffers, walk-ins, repeat visits.
Clinic/practitioner	Provider schedules, consultation types, privacy-conscious workflow; avoid unnecessary health data.
Tutor	Recurring lessons, student schedule, session completion.
Mechanic/repair	Vehicle or item type, estimated duration, drop-off/pick-up notes.
Home services	Service area, visit windows, travel buffers, access instructions.
Photographer	Long booking blocks, package selection, location, preparation instructions.
Fitness/wellness	Session packs, recurring sessions, class capacity where supported.
Consultant	Session duration, preparation notes, online meeting link.

5. Customer history: recommended screen design

The owner dashboard should include a dedicated **Customers** section. The list should show name, phone, last visit, next booking, completed visits, and a compact status. Opening a profile should show a summary at the top, then tabs or sections for Timeline, Notes, Upcoming, and Past appointments.

The appointment timeline should be chronological, scannable, and filterable by status or date. Each entry should expose only the actions appropriate to the user's role. Include an obvious “Book again” action that rechecks current availability rather than blindly duplicating the old time.

A new booking should link to a customer record when identity can be matched confidently. Use business_id + normalized phone as a practical lookup key, but do not assume a shared phone always means the same person. Offer confirmation when names conflict or family members share a number.

Suggested empty state: “Your customer history will appear here after your first booking.” Provide a clear action to add a customer or create a booking—never show fake customer data in the live product.

6. Reminder and notification workflows

Event	Customer notification	Owner visibility
Booking confirmed	Confirmation with service, local date/time, business details, and manage-booking link/reference.	Booking confirmed; notification status shown.
Upcoming appointment	Configured reminder(s), such as 24h and 2h before start.	Scheduled/sent/failed status; failure alert.
Owner approves request	Approval confirmation and appointment details.	Approval recorded.
Reschedule	Updated date/time; invalidate old reminder jobs and schedule new ones.	Change visible in timeline/audit.
Cancellation	Cancellation confirmation and next steps if applicable.	Slot becomes available; waitlist may be checked.
Reminder fails	Retry safely where appropriate; do not send endless duplicates.	Show failure reason/category and fallback action.
Waitlist slot opens	Time-limited offer if waitlist is enabled and consent allows.	Offer status and expiry visible.

Important distinction: a reminder being scheduled is not proof that a customer received it. Store the provider response and, where available, delivery receipts. Do not expose provider secrets to the browser.

7. UI/UX design direction

• Premium, calm, modern SaaS interface with responsive mobile-first layouts.

• Strong information hierarchy, readable typography, consistent spacing, clear icons, and accessible contrast/focus states.

• Owner dashboard: Today, Calendar, Bookings, Customers, Services, Reviews, Notifications, and Settings.

• Customer booking page: minimal steps, visible availability, transparent duration/price/policies, and reassuring confirmation.

• Customer profile: summary cards, timeline, quick rebook, contact action, and notes with clear permissions.

• Notifications screen: upcoming, failed, retrying, and completed statuses; do not imply delivery without evidence.

• Include skeleton/loading, empty, error, retry, offline/degraded, and success states. Use restrained animation and respect reduced-motion preferences.

• Light/dark theme if time permits; prioritize clarity and accessibility over decorative effects.

8. Cloudflare-first architecture

Recommended stack for the hackathon: React + TypeScript frontend hosted on Cloudflare Pages or Workers static assets; Cloudflare Workers for API/business logic; Cloudflare D1 for relational data; Cloudflare R2 for optional uploaded images; Cloudflare Queues and/or scheduled Workers for background notification jobs, depending on current platform capabilities and account limits.

• Keep authentication, tenant authorization, input validation, booking availability, booking writes, reminder scheduling, and notification state on the server.

• Use D1-supported transactional/conditional-write strategies and test simultaneous bookings. Never rely only on disabling a button in the UI.

• Store timestamps as UTC instants and retain the business IANA timezone for display and reminder calculation.

• For reminders, store durable jobs with due_at, status, attempt_count, idempotency_key, provider_message_id, and last_error_category.

• Scheduled Worker finds due jobs and enqueues work; worker sends through a configured provider, persists result, and retries transient failures with backoff.

• Use environment bindings/secrets for API keys; never commit secrets or put them in frontend environment variables.

• Do not promise native SMS/email/automated WhatsApp delivery without integrating and configuring an appropriate provider. Keep provider adapters swappable.

• Verify current Cloudflare service limits, D1 transaction semantics, Queue availability, and deployment settings before implementation; platform features and quotas can change.

9. Suggested data model

Entity	Key fields / purpose
businesses	id, owner_user_id, name, slug, timezone, contact settings, reminder defaults.
users / memberships	user id, business_id, role, permissions, session/auth metadata.
services	id, business_id, name, duration_minutes, buffer, price_minor_units, active.
staff/resources	id, business_id, display name, availability and service eligibility.
customers	id, business_id, name, normalized_phone, optional email, consent/preferences, created_at, updated_at.
bookings	id, business_id, customer_id, service_id, staff/resource, start_at_utc, end_at_utc, timezone, source, status, price sna
booking_events	booking_id, event_type, actor, old/new values, timestamp; audit timeline.
customer_notes	id, business_id, customer_id, author_user_id, body, visibility/sensitivity label, created_at.
notification_jobs	id, business_id, booking_id, customer_id, channel, template_key, due_at, status, idempotency_key, attempts, provid
notification_preferences	customer_id, channel, consent/status, updated_at; honor applicable policy.
reviews	booking_id, business_id, customer_id, rating, text, moderation status, owner reply.

Use tenant-scoped foreign keys and authorization checks. Normalize phone numbers consistently, but preserve a safe display version. Store money in integer minor units (paise in India) and use explicit currency. Avoid collecting unnecessary personal or sensitive data.

10. Booking and reminder business rules

• Validate service, staff/resource eligibility, opening hours, buffers, holidays, and booking policies on the server.

• Prevent conflicting active bookings with a reliable server-side write strategy; test two customers attempting the same slot simultaneously.

• Use a stable booking ID/reference and idempotency for create, cancel, reschedule, and notification operations.

• Calculate reminders in the business timezone, then persist an exact UTC due time. Handle daylight-saving changes for businesses in affected regions.

• When a booking is rescheduled or cancelled, mark prior reminder jobs as cancelled/superseded and create the correct replacement jobs.

• Do not schedule reminders for pending requests unless the business explicitly wants a pending-request reminder.

• Use configurable lead times and quiet hours; if a reminder time has already passed, follow a documented rule rather than sending unexpectedly.

• Never treat a provider's API acceptance as guaranteed delivery. Track accepted, delivered/read (if available), failed, and unknown separately.

11. Security, privacy, and reliability

• Strict business tenant isolation on every read and write; server-enforced role permissions.

• Secure authentication/session handling, validation, rate limits, CSRF protections where relevant, and safe logging.

• Limit customer history to authorized business users; restrict notes and exports by role.

• Obtain and record appropriate consent for outbound communications; provide opt-out/preference handling where required by the chosen channel and jurisdiction.

• Keep message content minimal; avoid sensitive details in lock-screen notifications. For clinics, use neutral wording where appropriate.

• Do not store passwords, provider secrets, or payment credentials in plain text.

• Audit changes to customer records, appointment status, notes, and notification settings.

• Define retention/deletion behavior and avoid deleting records needed for lawful accounting obligations without an appropriate policy.

• Monitor job backlog, failures, repeated retries, and provider rate limits. Provide a manual fallback if automation fails.

• Backups, migrations, restore checks, error monitoring, and a basic incident checklist before production use.

12. Analytics and success metrics

• Booking completion rate and median time to book.

• No-show rate before/after reminders, measured only after sufficient real data exists.

• Reminder success/failure rates by channel and provider; distinguish sent from delivered.

• Repeat-customer rate, time since last visit, and rebooking conversion.

• Schedule utilization, cancellations, recovered waitlist slots, and owner time spent managing bookings.

• Public page conversion and booking-source performance. Use opt-in/consent-respecting analytics and avoid exposing customer-level details unnecessarily.

13. Phased implementation roadmap

Phase	Deliverable	Priority
0. Foundation	Design system, routing, Cloudflare environment, database migrations, auth skeleton, error/loading states.	Must
1. Business setup	Business profile, service catalog, staff/resources, hours, holidays, timezone.	Must
2. Booking engine	Public URL, availability, customer form, server-side conflict prevention, confirmation, cancellation/reschedule.	Must
3. Owner operations	Calendar, booking list, Quick Add, source labels, walk-ins, statuses.	Must
4. Customer history	Customer directory, profile, phone matching, timeline, notes, quick rebook, duplicate handling.	Must
5. Reminder notifications	Durable reminder jobs, configurable schedule, one real configured channel, statuses, retries, failure UI, manual fallback.	Must for the planned scope; keep pr
6. Trust and sharing	QR/link sharing, ratings/reviews, owner replies, moderation basics.	Should
7. Reliability polish	Security pass, accessibility, tests, migrations/backups, monitoring, demo data clearly labeled.	Must before demo/release
Later	Payments, marketplace, advanced multi-location, official WhatsApp automation, deep offline sync, advanced analytics.	Defer

14. Hackathon cut line and acceptance criteria

Given limited time, finish one coherent end-to-end experience rather than many half-built modules. The strongest demo scope is: one business preset, service setup, public booking link, reliable booking conflict prevention, owner calendar, manual Quick Add, customer directory/history, reminder schedule, one actually configured notification channel or clearly labelled demo-mode notification, and polished sharing.

• A customer can book without installing an app and sees a real confirmation only after the booking is saved.

• A returning customer is matched safely and their past/upcoming appointments appear in the correct business profile.

• A staff member can open a booking and navigate to that customer's full timeline.

• A customer reminder is scheduled at the configured lead time; rescheduling/cancelling prevents stale reminders.

• Notification failures are visible, retry safely, and offer a manual fallback.

• Two concurrent attempts for one slot do not both become confirmed.

• No cross-business customer/booking access is possible.

• Mobile layout, accessibility, empty/error states, and demo data are checked.

15. Risks and scope boundaries

• External notification channels require provider credentials, sender setup, consent/template compliance, and possibly cost or approval. For the demo, configure one provider or show a clearly labelled simulated mode; never misrepresent simulated sends as real.

• Reminder delivery depends on third-party systems and customer contact details; design for failure and provide manual fallback.

• Phone-number matching can confuse family/shared numbers; show possible matches and let the owner confirm.

• Customer histories can contain personal information; collect the minimum needed and apply access controls and retention rules.

• Avoid building a general marketplace, payment platform, or complex CRM before the core booking/history/reminder journey works.

16. Demo script and final product promise

• Open the public business booking link on a phone-sized screen and book a service.

• Show the booking in the owner calendar and open the linked customer profile.

• Book again using the same customer's phone number and demonstrate history continuity.

• Open the customer timeline and use Quick Rebook.

• Show the reminder due time, message template, delivery/job status, and cancellation/reschedule behavior.

• Demonstrate a failed notification with a safe manual fallback; make it clear whether the channel is real or simulated.

• Share the public booking URL/QR and explain how the business centralizes website, WhatsApp, calls, and walk-ins.

Final product promise: “Wherever a customer discovers a business, they can book directly—and the business manages every appointment, customer relationship, and reminder in one place.”